

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

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Abstract

001 Long-horizon video generation is typically evaluated
002 with whole-frame metrics that reward motion and tem-
003 poral consistency. For fixed-camera nature scenes this
004 creates an ambiguity: motion in water, fire, smoke, or
005 rain is desirable, whereas motion of the background is
006 a generation error. A model can thus score highly on
007 motion while drifting, or on consistency while its flow
008 stagnates. We introduce SNF-Bench (Static-Natural
009 Flow), an evaluation framework for long-horizon fixed-
010 camera nature-flow generation that separates static
011 support from dynamic flow across text-to-video and
012 image-to-video tracks at 60–120 s horizons. Its proto-
013 col measures three factors: Static Fidelity, Flow Per-
014 sistence (retention jointly with absolute magnitude),
015 and Drift Leakage. Every measurement is validated
016 mechanistically: we inject global translation, rotation,
017 and scale drift and progressive motion attenuation at
018 known severity into real fixed-camera clips, and require
019 each factor to respond in its expected direction and
020 to remain selective against unrelated corruptions. Au-
021 diting publicly released long-horizon generators under
022 their intended configurations, we find that whole-frame
023 motion and static-region drift induce near-opposite or-
024 derings of the same outputs, and that the same con-
025 trolled drift which our static-fidelity factors detect in-
026 creases a widely used whole-frame motion score. SNF-
027 Bench evaluates where motion occurs and whether it
028 persists—not whether fluid dynamics are physically re-
029 alistic.

030 1. Introduction

031 Autoregressive and streaming video diffusion models
032 now extend generation from seconds to minute-scale
033 horizons. Evaluation, however, has not kept pace:
034 small temporal errors that are invisible in short clips
035 accumulate over long rollouts into background dis-

placement, color drift, repeated textures, or motion
decay, and whole-frame metrics summarize these qual-
itatively different failures into a single score.

The problem is sharpest in fixed-camera nature-flow
generation. In a static-camera video of a river, the wa-
ter should move while the riverbank, rocks, and sur-
rounding landscape remain stationary. The same sep-
aration governs waterfalls, ocean waves, rainfall, fire,
smoke, and wind-driven vegetation: some regions form
static support that must stay structurally fixed, while
others carry dynamic flow that must persist. Whole-
frame motion measures reward optical flow regardless
of its source—crediting background drift as readily
as genuine fluid motion—while temporal-consistency
measures can favor videos whose stability comes from
suppressing the very flow they should sustain. Two op-
posite failures, static drift and flow decay, are thereby
conflated, and increasingly so at long horizons, where
accumulated background motion can dominate a se-
quence’s apparent dynamics.

This ambiguity spans conditioning modalities. In
text-to-video (T2V) generation, the model must estab-
lish static support and preserve it without a visual ref-
erence; in image-to-video (I2V) generation, the input
image defines that support explicitly. Both tasks pose
the same fixed-camera question: does motion persist
where it is intended, without leaking into regions that
should remain static?

We introduce SNF-Bench (Static-Natural Flow), an
evaluation framework built around this static-dynamic
decomposition, with category-balanced T2V and I2V
tracks evaluated primarily at 60- and 120-second hori-
zons (shorter and ultra-long horizons serve as diagnos-
tics). Evaluation is organized into three factors. Static
Fidelity measures geometric displacement and resid-
ual optical flow inside nominally stationary regions; for
I2V, it additionally compares static regions against the
shared input image. Flow Persistence jointly reports
the retention of dynamic-region motion over time and
its absolute drift-compensated magnitude, so a persis-

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076 tence ratio cannot look strong merely because motion
077 is uniformly weak. Drift Leakage estimates how much
078 apparent dynamic-region motion is explained by global
079 background displacement.

080 A benchmark is only as trustworthy as its measure-
081 ments, so we validate the protocol independently of
082 any model comparison. Controlled perturbations inject
083 known translation, rotation, scale and color drift, mo-
084 tion attenuation, temporal repetition, and generation-
085 like corruptions into fixed-camera videos, testing that
086 each metric responds selectively and monotonically to
087 the failure it targets. We calibrate the drift-leakage
088 estimate on sequences with known mixtures of global
089 displacement and local motion, quantify region-mask
090 agreement, and verify that conclusions are stable un-
091 der mask erosion and dilation.

092 We then audit public long-horizon generators un-
093 der two explicitly separated settings: native-capability
094 evaluation, which runs each method in its intended
095 configuration and records its supported horizon, block
096 size, denoising schedule, and memory policy; and a
097 secondary matched stress test under a common fixed-
098 camera protocol. The analysis reveals cases where con-
099 ventional metrics and SNF-Bench interpret the same
100 outputs differently—high whole-frame motion coexist-
101 ing with substantial estimated drift leakage, and high
102 stability coexisting with decaying flow. We do not ar-
103 gue that existing metrics are wrong; we argue that
104 whole-frame aggregation does not represent a spatial
105 distinction that this setting requires. Nor does SNF-
106 Bench measure physical realism: a video may be drift-
107 free and persistent yet physically implausible. Our goal
108 is narrower and complementary—making the spatial
109 origin and temporal persistence of motion explicit.

110 Our contributions are:

- 111 1. SNF-Bench, a controlled evaluation framework for
112 long-horizon fixed-camera nature-flow generation,
113 with T2V and I2V tracks unified by a shared de-
114 composition of static support and dynamic flow.
- 115 2. A drift-motion evaluation protocol separately mea-
116 suring static fidelity, flow persistence with absolute
117 magnitude, and drift leakage.
- 118 3. Controlled validation of every measurement via syn-
119 thetic and generation-like perturbations, including
120 drift-leakage calibration and mask-robustness anal-
121 ysis.
- 122 4. A public-model audit under native and matched
123 configurations, showing that the fixed-camera de-
124 composition materially changes the interpretation
125 and ranking of long-horizon behavior relative to
126 generic metrics.

2. Related Work

Video generation evaluation. General-purpose bench-
marks assess complementary aspects of visual quality,
temporal consistency, motion, and semantic alignment;
VBench [2], for example, decomposes quality into di-
mensions including Background Consistency, Motion
Smoothness, Temporal Flickering, and Dynamic De-
gree. These suites enable broad comparison across het-
erogeneous generators, but their measurements are pri-
marily whole-frame or sequence-level. In fixed-camera
nature-flow scenes, whole-frame aggregation cannot re-
veal whether measured motion originates in intended
dynamic regions or in displacement of nominally static
structure. SNF-Bench complements such suites by
making the spatial origin of motion explicit; it replaces
none of them.

Long-horizon autoregressive generation. Autoregres-
sive and streaming generators extend synthesis beyond
short bidirectional clips by producing frames or tem-
poral chunks causally, including CausVid [10], Self-
Forcing [1], Infinite-Forcing [3], Rolling-Forcing [5],
Reward-Forcing [6], LongLive [8], and Causal-
Forcing [11]. These systems differ substantially in tem-
poral block size, training objective, memory and cache
policy, motion prior, and horizon-extension mecha-
nism [9]. Their evaluation, however, typically re-
lies on generic video metrics or method-specific qual-
itative analysis. The deployment diversity motivates
SNF-Bench’s two-setting audit—native capability un-
der each released configuration, and a separately re-
ported matched stress test—while the evaluation gap
motivates its method-agnostic question: does static
support remain fixed while dynamic flow persists?

Image dynamics and nature-flow animation. Work on
animating still images has long modeled natural motion
explicitly: controllable fluid animation separates static
image structure from specified fluid motion [7], and
Generative Image Dynamics models long-term scene
motion through learned trajectory representations [4].
These works affirm that static support and dynamic
content are distinct modeling targets—the same dis-
tinction SNF-Bench elevates to an evaluation principle
for modern long-horizon generators.

Image-to-video faithfulness. I2V generation adds a
reference signal: the source image specifies the ini-
tial scene. Existing I2V evaluation commonly mea-
sures source or subject consistency alongside temporal
quality, but high source similarity does not determine
whether motion remains confined to regions expected

176 to move, nor whether stability was achieved by sup-
177 pressing motion. SNF-I2V applies the same static-
178 dynamic decomposition as SNF-T2V, adding source-
179 referenced static-region fidelity as one complementary
180 factor.

181 3. SNF-Bench Design

182 SNF-Bench evaluates long-horizon generation under a
183 deliberately constrained setting: a fixed camera observ-
184 ing a scene containing both stationary structure and
185 persistent natural motion. This isolates two failure
186 modes that are difficult to separate in unconstrained
187 video: static drift, where nominally stationary struc-
188 ture moves over time, and flow decay, where intended
189 dynamics weaken or disappear. Every sequence is
190 treated as a composition of two spatial roles,

$$191 \quad \Omega_{\text{static}} \quad \text{and} \quad \Omega_{\text{flow}}, \quad (1)$$

192 where Ω_{static} denotes regions expected to remain ge-
193 ometrically stationary and Ω_{flow} regions expected to
194 exhibit sustained motion. This partition defines an
195 evaluation role, not a semantic ontology: a riverbank
196 is static support while the river surface is flow; build-
197 ings are static support in a rainfall scene while falling
198 rain is flow. SNF-Bench does not require dynamics in-
199 side Ω_{flow} to follow any particular physical model—it
200 measures whether motion stays localized to appropriate
201 regions and persists over time without contaminating
202 static support.

203 3.1. T2V and I2V Tracks

204 SNF-T2V. The model receives only a text prompt de-
205 scribing a fixed-camera nature-flow scene. With no vi-
206 sual reference, the opening of the sequence establishes
207 the scene’s static support; evaluation asks whether that
208 support remains stable over the rollout while flow re-
209 mains active. All methods receive identical prompts
210 per item.

211 SNF-I2V. The model receives a shared source image
212 and a text prompt describing intended motion. The
213 image specifies initial scene structure explicitly; the
214 same image, prompt, and static-region mask are used
215 for every method. The I2V track additionally permits
216 direct comparison between late frames and the input
217 image inside Ω_{static} .

218 The tracks differ only in how static support is
219 established—generated internally (T2V) or externally
220 specified (I2V)—and share one evaluation question:
221 does intended motion persist without inducing motion
222 in static regions?

3.2. Scene Categories and Horizons

The benchmark is category-balanced across representa-
tive fixed-camera natural processes—rivers and canals,
waterfalls, ocean waves, rainfall, snowfall, fire and lava,
smoke and clouds, and storm-driven motion—with cat-
egories fixed before the final sweep and all aggregates
macro-averaged across categories so no scene type dom-
inates. Four temporal regimes are used: 5s (sanity
check), 60s and 120s (primary evaluation), and 240s
(diagnostic stress test). Principal conclusions rest on
the 60s and 120s tiers. A method that cannot reach
a horizon under its native public configuration is re-
ported as unsupported rather than silently modified.

3.3. Method-Independent Region Masks

Region definitions must not depend on the tempo-
ral behavior of the model under evaluation: deriv-
ing masks from a generated video’s own motion field
would be circular, since background drift could itself
be classified as evidence that the affected region is dy-
namic. SNF-Bench therefore excludes every evaluated
model’s motion field from its own scoring mask, using
the strongest mask protocol each track admits.

Derivation. Both tracks use the same automatic pro-
cedure. For each generated sequence we compute
optical-flow magnitude over an early window—the
leading 12% of sampled frame pairs—average it, and
threshold it by Otsu’s method into a dynamic region
and its complement. The static region is then shrunk
by a fixed morphological erosion and a 4% frame border
is discarded, which together act as the ignored transi-
tion band: boundary pixels, where the partition is least
certain and flow is least reliable, contribute to neither
region.

Why the early window. Deriving a mask from a gener-
ated video’s whole motion field would be circular, since
accumulated background drift would itself be read as
evidence that the affected region is dynamic. Restrict-
ing the estimate to the early window breaks that cir-
cularity in the direction that matters: at 60s the mask
is fixed from the first 7.1s of the rollout, before long-
horizon drift has accumulated, and the metrics are then
evaluated over the remainder. The partition there-
fore reflects each method’s spatial layout—which is un-
avoidable for T2V, where every method authors its own
scene—but is not defined by the temporal failure being
measured.

What this protocol does not provide. We state the
limitation plainly, because it bounds the strength of ev-

271	ery conclusion drawn from these masks. The partition	321
272	is obtained automatically rather than from human an-	322
273	notation, so no inter-annotator agreement is reported,	323
274	and for I2V the mask is derived from each generated	324
275	sequence rather than once from the shared source im-	325
276	age. The early window is a weaker guarantee than	326
277	a first-frame annotation: a method that drifts within	327
278	the first 12% of its rollout can influence its own parti-	328
279	tion. We therefore treat masks as measured quantities	329
280	rather than ground truth and quantify the consequence	330
281	directly—Section 5 recomputes every measurement un-	
282	der systematic erosion and dilation of the partition and	
283	reports the resulting stability. A human-verified, first-	
284	frame-only annotation protocol is the natural strength-	
285	ening of this component and is the principal item we	
286	leave to future work.	
287	Layered content. A two-way partition cannot repre-	
288	sent scenes in which dynamic content transits static	
289	support: rain and snow pass in front of buildings, so	
290	the same pixels are simultaneously support and in-	
291	tended motion. SNF-Bench assigns such pixels by flow	
292	magnitude like any others, which means static-fidelity	
293	measurements in precipitation categories carry an addi-	
294	tional, unmodelled contribution. These categories are	
295	reported rather than dropped, and per-category results	
296	(Section 7.3) make the affected subset explicit.	
297	Masks are stored at reference resolution and resam-	
298	pled to each video with nearest-neighbor interpola-	
299	tion; a small ignored transition band around static-flow	
300	boundaries prevents boundary pixels from dominating	
301	either region, with its width frozen in the benchmark	
302	specification and included in the sensitivity analysis.	
303	3.4. Frozen Manifest and Failure Handling	
304	SNF-Bench is defined by a versioned manifest frozen	
305	before the final public-model sweep, recording per item:	
306	identifier and category, conditioning modality, exact	
307	prompt, source-image identifier (I2V), region mask,	
308	target horizons, resolution and frame-rate metadata,	
309	and provenance information. After freezing, items are	
310	never removed because a method performs poorly; cor-	
311	rections require a new benchmark version. A failure	
312	taxonomy—unsupported horizon, out-of-memory, run-	
313	time crash, numerical corruption, invalid output—is	
314	likewise frozen in advance; failures are reported along-	
315	side successful generations, never silently excluded.	
316	3.5. Native Capability and Matched Stress-Test Set-	
317	tings	
318	Public generators are evaluated under two explicitly	
319	separated settings, defined here so the audit protocol	
320	cannot appear post hoc.	
	Native capability (primary). Each method runs under	321
	its intended public inference configuration. We record	322
	checkpoint, temporal block size, denoising schedule,	323
	guidance, cache and memory policy, resolution, frame	324
	rate, and the longest horizon the released implemen-	325
	tation supports. Long-video generators differ substan-	326
	tially in temporal chunking and cache design; SNF-	327
	Bench treats these differences as part of native capa-	328
	bility rather than assuming any common wrapper is	329
	neutral for every checkpoint.	330
	Matched stress test (secondary). Compatible check-	331
	points are additionally evaluated under a common	332
	fixed-camera long-horizon protocol. This setting asks	333
	how behavior changes under a shared deployment	334
	regime; it is not interpreted as a native ranking, and	335
	its results are reported separately.	336
	3.6. Aggregation and Uncertainty	337
	Metrics are computed per item and macro-averaged	338
	across categories. Uncertainty is estimated with	339
	prompt-level paired bootstrap resampling; we report	340
	95% confidence intervals and treat methods whose	341
	intervals substantially overlap as statistically indis-	342
	tinguishable rather than assigning definitive ordinal	343
	ranks.	344
	4. Drift-Motion Evaluation Protocol	345
	SNF-Bench evaluates fixed-camera generation along	346
	three complementary factors: Static Fidelity, Flow	347
	Persistence, and Drift Leakage. No individual score is	348
	sufficient: low background motion can be achieved by	349
	freezing the entire video, and high foreground motion	350
	can be produced by drifting the entire scene. We there-	351
	fore always report static-region stability jointly with	352
	both the relative persistence and absolute magnitude	353
	of dynamic-region motion.	354
	Let I_t denote frame t of a T -frame sequence, Ω_{static}	355
	and Ω_{flow} the fixed region masks of Sec. 3.3, and $\mathbf{u}_t(\mathbf{x})$	356
	the optical flow from I_t to I_{t+1} at pixel $\mathbf{x}$. Each se-	357
	quence defines an early window $\mathcal{E}$ and a late window	358
	$\mathcal{L}$ of equal duration; $\mathcal{E}$ begins after a short warm-up	359
	interval so that initial generation transients do not de-	360
	fine the motion reference. Window extents are fixed in	361
	the benchmark specification.	362
	4.1. Static Fidelity	363
	Feature-aligned Background Drift (fBD). We extract	364
	repeatable local features inside Ω_{static} from the first	365
	frame and match them against each frame I_t in the	366
	late window, applying robust geometric filtering to re-	367
	ject outlier correspondences. For the resulting matched	368

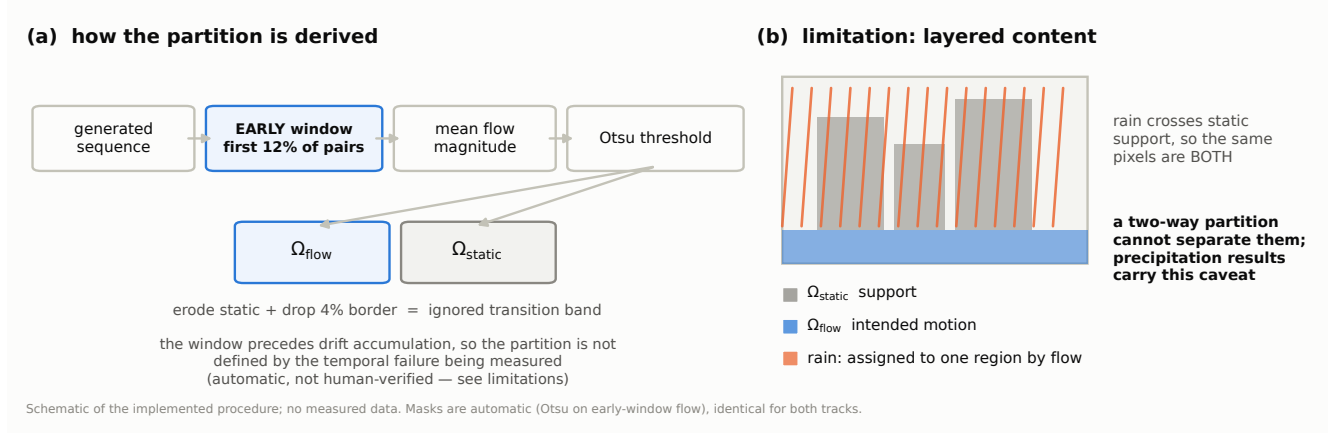


Figure 1. Region-partition protocol. (a) The partition is obtained automatically and identically for both tracks: mean optical-flow magnitude over the leading 12% of sampled frame pairs is thresholded by Otsu’s method, the static region is eroded and a 4% border discarded to form an ignored transition band. Restricting the estimate to the early window is what prevents accumulated drift from defining the very region against which drift is measured. (b) A two-way partition cannot represent layered content: rain and snow transit static support, so those pixels are assigned to one region by flow magnitude and precipitation results carry an additional unmodelled contribution.

369 pairs $\mathcal{M}_t$,

$$370 \quad D_{\text{feat}}(t) = \text{median}_{(p,q) \in \mathcal{M}_t} \|p - q\|_2, \\ 371 \quad \text{fBD} = \frac{100}{d} \cdot \frac{1}{|\mathcal{L}|} \sum_{t \in \mathcal{L}} D_{\text{feat}}(t) \quad (\downarrow), \quad (2)$$

372 where d is the frame diagonal, so fBD is expressed as
373 a percentage of image size. fBD captures accumulated
374 geometric displacement of persistent structure and is
375 intentionally insensitive to fine-scale flicker that moves
376 no structure.

377 Normalized Background Flow (NBF). Complementing
378 accumulated drift, NBF measures instantaneous
379 motion energy inside the static mask:

$$380 \quad \text{NBF} = \frac{10^3}{W \Delta t} \cdot \frac{1}{T-1} \sum_{t=1}^{T-1} \frac{1}{|\Omega_{\text{static}}|} \sum_{\mathbf{x} \in \Omega_{\text{static}}} \|\mathbf{u}_t(\mathbf{x})\|_2 \quad (\downarrow), \quad (3)$$

381 normalized by frame width W for resolution invariance
382 and by the inter-frame interval Δt for frame-rate in-
383 variance, giving units of 10^{-3} frame widths per sec-
384 ond. Here Δt is the interval of the sampled pairs actu-
385 ally used for flow estimation: methods are evaluated at
386 their native frame rates, which differ across the roster,
387 and the evaluation pipeline subsamples each clip to a
388 common temporal rate before measurement. Expressing
389 NBF per unit time rather than per frame makes the
390 quantity independent of both the native rate and the
391 sampling policy, so the measure remains comparable if
392 either changes.

NBF is a normalized magnitude, not a ratio: we
deliberately do not divide by whole-frame flow, since
that would couple background leakage to foreground
behavior and inflate the measure for methods whose
dynamic motion decays.

I2V source fidelity. For SNF-I2V, we additionally re-
port feature-space similarity between the source image
and late frames restricted to Ω_{static} , measuring whether
static regions preserve source appearance rather than
merely remaining internally stationary.

4.2. Flow Persistence

Global-motion compensation. Coherent scene dis-
placement is removed before dynamic-region motion is
evaluated. A single translation vector is insufficient:
under rotation or zoom the static-region flow field is
not constant, and its component-wise median is near
zero for a centered rotation, so a translation-only es-
timate would leave exactly the corruptions Sec. 5 in-
jects uncompensated. We therefore estimate a robust
global similarity transform $T_t(\mathbf{x}) = s_t R_t \mathbf{x} + \mathbf{b}_t$ from
static-region correspondences $\mathbf{x} \mapsto \mathbf{x} + \mathbf{u}_t(\mathbf{x})$ under
RANSAC, define the induced global displacement field
 $\mathbf{d}_t(\mathbf{x}) = T_t(\mathbf{x}) - \mathbf{x}$, and compensate pointwise:

$$\hat{\mathbf{u}}_t(\mathbf{x}) = \mathbf{u}_t(\mathbf{x}) - \mathbf{d}_t(\mathbf{x}). \quad (4)$$

We restrict to similarity rather than unrestricted
affine because fixed-camera generation failures plausi-
bly manifest as translation, slight rotation and zoom,

whereas affine shear can absorb legitimate local deformation and over-correct. When correspondences are insufficient or the robust fit fails, we fall back to median translation and record the fallback per video; fallback rates are reported per category.

Motion-Compensated Foreground Flow (MCFF). For a temporal window $\mathcal{W}$,

$$\text{MCFF}(\mathcal{W}) = \frac{1}{|\mathcal{W}|} \sum_{t \in \mathcal{W}} \frac{1}{|\Omega_{\text{flow}}|} \sum_{\mathbf{x} \in \Omega_{\text{flow}}} \|\tilde{\mathbf{u}}_t(\mathbf{x})\|_2 \quad (\uparrow). \quad (5)$$

We report $\text{MCFF}(\mathcal{E})$ and $\text{MCFF}(\mathcal{L})$, so a method with uniformly negligible motion cannot appear strong through relative measures alone.

Flow Persistence (FP).

$$\text{FP} = \min\left(\frac{\text{MCFF}(\mathcal{L})}{\text{MCFF}(\mathcal{E}) + \epsilon}, c\right) \quad (\uparrow), \quad (6)$$

clipped at a fixed constant c to prevent unstable ratios when early motion is small. FP near one indicates retained motion; substantially smaller values indicate decay; values above one are not automatically better. FP is never reported without its MCFF magnitudes.

4.3. Drift Leakage and Attenuation

Whole-frame metrics cannot determine whether apparent dynamic-region motion arises from intended flow or from coherent displacement of the whole scene. We report two distinct diagnostics for this, because they answer different questions and neither subsumes the other. Let F_{static} denote mean flow magnitude over Ω_{static} , and F_{raw} , F_{comp} the mean raw and compensated flow magnitude over Ω_{flow} , all in the late window; the late window is used because drift accumulates over the rollout.

Drift Leakage Ratio (DLR).

$$\text{DLR} = \frac{F_{\text{static}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow) \quad (7)$$

measures background-flow magnitude relative to raw dynamic-region flow—that is, where the sequence’s motion energy is located. It is deliberately not normalized to a bounded range and is not a fraction: $\text{DLR} > 1$ indicates that static support moves more than the region intended to move, a regime that occurs in practice.

Drift Attenuation Ratio (DAR).

$$\text{DAR} = 1 - \frac{F_{\text{comp}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow) \quad (8)$$

measures the signed relative change in dynamic-region flow after global-motion compensation—that is, how much of the apparent dynamic motion is explained by a global displacement model. Near-zero values mean compensation removes little, so motion is genuinely local; large values mean apparent motion is largely accounted for by scene displacement. Headline values are reported as $\text{clip}(\text{DAR}, 0, 1)$, while the signed quantity is stored and used for validation. Negative values are informative rather than anomalous: where local flow opposes the estimated global field, compensation can increase measured magnitude. We report the incidence of negative values per method rather than suppressing it.

Interpretation. DAR is a calibrated proxy for global-motion contamination, not a causal decomposition of motion into drift and non-drift components; Sec. 5 calibrates it against sequences with known injected displacement. Accordingly we phrase results as “global compensation removes $X\%$ of measured dynamic-region flow under this estimator,” never as “ $X\%$ of the motion is drift.” Read jointly, the pair is more informative than either alone: high DLR with moderate DAR indicates large static-region motion energy that a rigid global model does not explain, implicating non-rigid instability such as warping rather than uniform scene translation.

4.4. Joint Interpretation

The factors are interpreted jointly: low fBD/NBF with low MCFF indicates an over-stabilized, nearly frozen video; high MCFF with high NBF, DLR or DAR indicates motion contaminated by displacement; high FP with negligible MCFF indicates persistent but trivial motion. The desired regime is low static drift, non-trivial compensated motion, and high retention. SNF-Bench deliberately provides no single scalar leaderboard score—collapsing the factors would recreate the exact ambiguity the benchmark exposes. Methods are instead compared per factor under the paired uncertainty analysis of Sec. 3.6.

Implementation convention. Videos are normalized to the benchmark evaluation resolution before measurement, and temporally subsampled to a common flow-estimation rate; no frame interpolation is performed, since synthesizing intermediate frames would alter the very flow field being measured. Residual frame-rate dependence is handled analytically by the Δt term in NBF, so the reported quantity is invariant to both the native rate and the subsampling policy. The optical-flow estimator, local-feature extractor, robust filtering

506 thresholds, warm-up interval, window extents, clip con-
507 stant c , boundary transition band, and ϵ are fixed glob-
508 ally before the final sweep and released with the evalu-
509 ation code; the identical implementation is applied un-
510 changed to every method. Per-video native frame rate,
511 resolution, and global-motion compensation mode are
512 released alongside the scores.

513 5. Controlled Validation

514 Before SNF-Bench is used to compare generators, we
515 validate that its measurements respond to the failure
516 modes they target. A benchmark is not justified merely
517 because its metrics produce different rankings; each
518 metric must change selectively, monotonically, and in
519 the expected direction under known corruptions. We
520 therefore construct controlled variants of fixed-camera
521 reference clips with known corruption type and sever-
522 ity. Validation is performed independently of the audit
523 in Sec. 6: perturbation strengths, masks, temporal win-
524 dows, and metric hyperparameters are fixed before the
525 final baseline results are inspected.

526 5.1. Mechanistic Perturbations

527 Given a reference sequence $V = \{I_t\}_{t=1}^T$, we generate
528 variants isolating individual components of the static-
529 dynamic decomposition.

530 Geometric drift. Temporally increasing global trans-
531 formations

$$532 I'_t(\mathbf{x}) = I_t(A_t \mathbf{x} + \mathbf{b}_t) \quad (9)$$

533 simulate camera-like background displacement, with
534 A_t (rotation, scale) and $\mathbf{b}_t$ (translation) growing pro-
535 gressively to match the accumulated nature of long-
536 horizon errors. Translation, rotation, and scale drift
537 are evaluated independently across severity levels.
538 Static-fidelity metrics should increase monotonically;
539 after compensation, MCFF and FP should remain com-
540 paratively stable, since the local dynamic process is un-
541 changed.

542 Photometric drift. Gradual low-frequency changes
543 in brightness, contrast, and color temperature pre-
544 serve geometry and motion. Geometric drift mea-
545 sures should remain comparatively insensitive; because
546 optical-flow estimation is not perfectly photometrically
547 invariant, we measure rather than assume the residual
548 sensitivity of flow-based quantities and report it as an
549 operating bound. For I2V, this perturbation also ver-
550 ifies that source-fidelity measurements detect appear-
551 ance change that displacement metrics cannot.

Motion attenuation and freezing. Motion inside Ω_{flow}
is progressively attenuated or frozen after predeter-
mined fractions of the sequence, with static support
untouched—known instances of decaying persistence
without background drift. MCFF should decrease with
attenuation and FP with late-stage decay; fBD and
NBF should remain approximately unchanged.

Temporal repetition. Late dynamic-region content is
replaced with short repeated cycles, producing per-
sistent apparent activity without genuine progression.
We include this control to verify that static-drift met-
rics are unaffected and to document explicitly that per-
sistence and magnitude cannot resolve repetition: a
looping flow can maintain both FP and MCFF. This
is a scope boundary of SNF-Bench, consistent with its
stated exclusion of motion naturalness.

515 5.2. Selectivity, Monotonicity, and Low-Motion Sta- 516 bility

517 For each perturbation family we quantify: mono-
518 tonic response—Spearman’s ρ between ordered sever-
519 ity $\lambda_1 < \dots < \lambda_K$ and metric response, with ex-
520 pected signs fixed in advance; selectivity—normalized
521 metric change under the target perturbation compared
522 against changes under unrelated perturbations (e.g.,
523 fBD should react strongly to translation but weakly to
524 freezing; MCFF the reverse); and low-motion stabil-
525 ity—explicit tests on low-amplitude and progressively
526 attenuated clips verifying that the clip constant c , to-
527 gether with mandatory joint reporting of early/late
528 MCFF, prevents small denominators from producing
529 misleading persistence conclusions.

530 5.3. Calibration of Drift Leakage and Attenuation

531 DLR and DAR are the most interpretive quantities in
532 SNF-Bench, so both are calibrated against sequences
533 with known global-motion contributions, and neither
534 is designated the primary drift diagnostic in advance
535 of this test. Starting from a reference clip with local
536 dynamic motion, we superimpose a controlled global
537 displacement on the entire frame—as genuine drift
538 manifests—so that the static-region estimate observes
539 the same displacement that contaminates Ω_{flow} . Vary-
540 ing injected magnitude relative to original foreground
541 flow yields clips with known drift-to-local-motion mix-
542 tures. With r_{inj} the known relative global-motion
543 contribution and $\hat{m}$ the estimate produced by metric
544 $m \in \{\text{DLR}, \text{DAR}\}$, we report

$$545 \rho_m = \rho_{\text{Spearman}}(r_{\text{inj}}, \hat{m}), \quad (10)$$

546 together with mean absolute calibration error, mono-
547 tonicity violations across injection levels, cross-

category variance, and—for DAR—the incidence of negative values. DAR is calibrated in its signed form, since clipping would conceal precisely the cases where compensation increases measured magnitude.

We do not expect exact recovery of a physical decomposition on arbitrary generated video; the claim established here is narrower—that these quantities behave as calibrated proxies whose values increase predictably as coherent displacement contributes more to apparent motion. A metric that fails monotonicity, proves unstable across categories, or depends materially on the flow backbone is reported as a supplementary diagnostic rather than a headline factor.

5.4. Mask Sensitivity

Region masks are treated as measured quantities, not ground truth. Because the partition is derived automatically (Sec. 3.3) rather than annotated, the relevant question is not inter-annotator agreement—which this release does not measure—but whether conclusions survive reasonable variation of the boundary. Every evaluation mask is therefore systematically eroded and dilated by fixed radii and all measurements recomputed, with the dynamic region varied complementarily so the test perturbs the boundary rather than the total measured area. We report the resulting response of each factor in Table 1; a factor whose value is dominated by boundary placement is not admitted as a headline quantity.

5.5. Validation Results

Figure 2 and Table 1 report the outcome. Injected translation produces the expected monotone increase in both static-fidelity factors, and progressive attenuation of dynamic-region content drives MCFF-L and FP down while leaving the static factors comparatively unchanged—the selectivity the decomposition requires. The comparison that motivates the benchmark appears in the same experiment: the identical injected background drift that our static-fidelity factors register also raises VBench Dynamic Degree, so a whole-frame motion score improves on precisely the corruption a fixed-camera evaluation should penalise.

Two results are reported because they qualify the framework rather than support it. fBD responds strongly and monotonically to translation but behaves erratically under pure rotation, where feature correspondences degrade; NBF carries the static-fidelity signal in that regime, which is why the two are reported jointly and why neither is used alone. Reported alongside them is the scope boundary of Sec. 5.1: looping dynamic content sustains both persistence and magnitude, so SNF-Bench cannot distinguish genuine pro-

perturbation	clips	fBD	NBF	MCFF-L	FP	DLR	DAR	VB-DD
photometric	5	-0.30	+0.90	+0.70	+0.60	-0.90	-0.90	+0.80
mask radius	5	+0.10	+1.00	-0.50	-1.00	+1.00	-0.90	—
translation	5	+1.00	+1.00	-0.90	-0.30	+1.00	-0.60	+0.90
rotation	5	-0.40	+0.20	-0.10	+0.70	-0.20	-0.10	+0.20
scale	5	+0.90	+0.50	-0.40	+0.00	-1.00	+0.80	+0.60
attenuation	5	-0.60	-1.00	+0.00	+0.00	+1.00	-1.00	-1.00
freeze	5	-1.00	-1.00	-1.00	-1.00	-0.40	+0.80	-1.00
repetition	5	+0.40	-0.20	+0.20	-0.40	+0.00	-0.40	-0.40

Table 1. Mechanistic validation. Spearman correlation between injected severity and each factor’s response, over controlled perturbations of real fixed-camera clips. Signs are fixed in advance by each definition. mask radius perturbs the region partition rather than the video, so near-zero entries indicate the desired insensitivity. Cells where a factor responds weakly or non-monotonically are reported rather than omitted.

gression from repetition, and does not claim to.

5.6. Validation Criteria

A metric is admitted to the public-model audit only if, under pre-specified criteria: (1) its response direction matches expectation under the target perturbation; (2) the response is monotonic across severity within statistical uncertainty; (3) unrelated perturbations produce substantially smaller changes; (4) ratio-based quantities remain interpretable under low-motion controls; and (5) conclusions are stable under erosion and dilation of the region partition. Metrics or configurations failing these checks are revised or excluded before the final sweep. The validation suite is thus an independent gate on the benchmark specification, not a post-hoc explanation of model rankings.

6. Public-Model Audit

We apply SNF-Bench to publicly available long-horizon video generators. The goal is not a single universal leaderboard, but a determination of how deployed systems occupy the static-fidelity–flow-persistence space under conditions both faithful to each released method and comparable under a common stress test. Two evaluation settings are therefore separated throughout and their scores are never merged.

6.1. Evaluated Models

The audit includes publicly released autoregressive or streaming generators with accessible checkpoints and reproducible inference code, frozen as a set before the baseline sweep. Unpublished or internal systems, and methods whose outputs cannot be independently reproduced, are excluded. The I2V track evaluates public models with a documented image-conditioned generation pathway. All checkpoints, code revisions, and

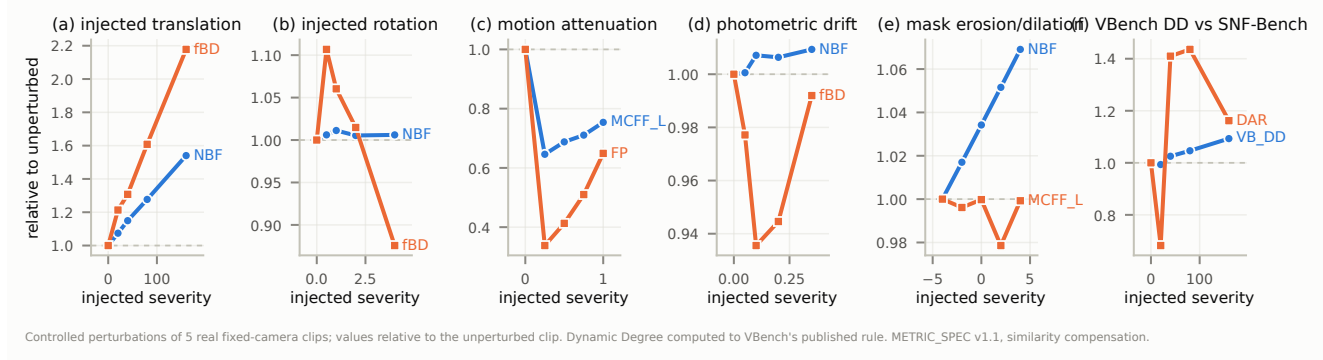


Figure 2. Mechanistic validation. Each panel injects a corruption of known kind and severity into real fixed-camera clips and plots each factor relative to its unperturbed value. Panel (f) puts VBench Dynamic Degree through the identical corruption as the SNF factors: the score rises with injected background drift. The perturbation is synthetic, so the comparison implicates no published method.

inference settings are recorded in the benchmark manifest.

6.2. Setting A: Native-Capability Evaluation

The primary audit runs each method in the deployment regime its public release intends, answering a capability question: what behavior does the released system exhibit when used as its authors intended? For every method we record: public checkpoint and code revision; conditioning mode; temporal block or chunk size; denoising-step schedule; guidance configuration; KV-cache, rolling-window, or memory policy; positional-encoding or horizon-extension mechanism; output resolution and frame rate; and the longest horizon supported without modifying the released temporal-generation policy. Operational limits are recorded alongside these settings (Sec. 6.7).

Method-specific differences in block size, schedule, and memory policy are preserved deliberately: they are part of the released system, not nuisance variables. Native results are therefore comparisons of deployed systems, not iso-compute comparisons of isolated checkpoints. A method that cannot reach a target horizon without changing a mechanism central to its released procedure is marked unsupported at that horizon; we never silently substitute its block size, cache policy, or schedule to populate a table entry. Crashes, numerical failures, and invalid outputs follow the frozen taxonomy of Sec. 3.4.

6.3. Setting B: Matched Long-Horizon Stress Test

Native evaluation cannot isolate how a checkpoint responds to a shared long-horizon regime, since released systems differ in chunking and budget. A second, explicitly secondary setting therefore tests compatible checkpoints under a common fixed-camera protocol,

answering a different question: how does the checkpoint behave under the same deployment constraint? The matched protocol fixes resolution, frame rate, horizon, prompt set, and evaluation windows; where technically compatible, the temporal generation budget is harmonized, and every departure from a method's native configuration is documented per checkpoint.

Matched scores are never used to redefine native capability or attributed to the original published system. They measure deployment sensitivity: whether conclusions about drift and persistence survive moving a checkpoint away from its intended temporal regime. This matters especially for autoregressive generators, whose behavior can depend strongly on framewise versus chunkwise rollout and cache structure. Native and matched results appear in separate tables, and disagreement between them is analyzed as an evaluation finding, never averaged away.

6.4. Common Inputs and Reproducibility Controls

Within each benchmark item, all methods receive identical semantic conditioning: the same frozen text prompt (T2V), or the same source image and prompt subject only to each implementation's required preprocessing (I2V). Prompts are never rewritten per method. The manifest fixes the seed set before the final sweep. Seed control guarantees reproducibility within each method; because implementations consume randomness differently, identical seeds do not induce matched noise draws across methods, and cross-method comparison relies on the paired-item design rather than paired noise. Methods that remain nondeterministic under seed control are documented and given additional repetitions for variance estimation. Generated videos are retained in native form; metric computation applies the common normalization of Sec. 4 unchanged

to every method.

6.5. Statistical Comparison

Scores are computed per benchmark item and macro-averaged across categories. Uncertainty is estimated by paired bootstrap resampling over items, preserving category membership; we report 95% confidence intervals and pairwise effect estimates. Exact ordinal ranks are avoided where uncertainty does not support them—methods whose paired differences are not reliably distinguishable are reported as one comparison group. Category-wise results are reported alongside aggregates: a method may preserve static geometry in rain scenes yet fail on broad ocean motion, and such category dependence is a benchmark result, not noise.

6.6. Comparison with Generic Video Metrics

To test whether the static–dynamic decomposition changes interpretation, the same outputs are additionally scored with established whole-frame metrics, and we compare the ordering they induce against the factor-level behavior exposed by SNF-Bench. The focus is interpretation change, not declaring existing metrics incorrect. In a representative case, a model receives a strong whole-frame motion score while exhibiting high NBF, DLR or DAR: the generic metric correctly confirms that substantial image motion exists, while SNF-Bench reveals that part of it occurs in regions expected to remain stationary. Conversely, strong temporal-consistency scores may coexist with weak late MCFF and low FP, indicating stability partly achieved through motion decay. Both cases motivate joint reporting rather than replacement.

6.7. Capability and Failure Reporting

For each system we report the operational boundary of its released implementation—unsupported horizons, failed generations, memory exhaustion, numerical corruption, invalid outputs—as rates against the frozen manifest, never removed after inspection. Coverage is counted from valid per-video measurements rather than from the presence of a results file, so an entry whose measurement run terminated early is reported as partial rather than complete.

6.8. Audit Questions

The audit is organized around four questions fixed before the final sweep: (1) static fidelity—which systems best preserve stationary regions as horizon increases; (2) motion persistence—which maintain non-trivial drift-compensated motion in Ω_{flow} from early to late rollout; (3) metric interpretation—when generic whole-frame metrics and SNF-Bench yield materially

Method	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
60s horizon ($n = 23$)							
CausVid	8.08	9.27	2.87	1.96	0.67	0.45	0.14
Self-Forcing	11.4	12.7	10.4	2.54	0.44	0.50	0.14
Infinite-Forcing	7.75	3.65	1.17	0.29	0.63	0.55	0.14
Rolling-Forcing	16.8	11.2	3.18	1.57	0.75	0.54	0.18
Reward-Forcing	14.8	8.51	3.81	0.80	0.49	0.60	0.17
LongLive	15.9	13.9	6.46	2.41	0.75	0.59	0.17
Causal-Forcing	22.4	87.6	9.84	7.21	0.83	0.94	0.43
120s horizon ($n = 6$)							
CausVid	6.15	3.42	0.36	0.52	1.06	0.41	0.13
Self-Forcing	21.2	6.80	4.85	0.82	0.85	0.75	0.15
Infinite-Forcing	16.1	2.60	0.50	0.32	0.64	0.57	0.21
Rolling-Forcing	23.8	10.6	3.33	0.77	0.38	0.82	0.22
Reward-Forcing	20.0	11.6	2.05	0.69	0.39	0.70	0.12
LongLive	22.5	9.66	5.10	1.19	0.81	0.84	0.23
Causal-Forcing	21.8	130	10.5	10.7	0.90	0.81	0.50

Table 2. SNF-Bench audit, T2V track, native configuration. All systems are public external models run under their own intended configuration. Our own systems are excluded by construction. DAR is reported clipped to $[0, 1]$; signed values are released. All systems are evaluated in the native setting. n is stated per horizon and is common to every row of that panel.

different conclusions about the same videos; and (4) deployment sensitivity—how stable those conclusions are between native and matched settings. Results in Sec. 7 are reported as answers to these questions, not as post-hoc narratives around any preferred model.

7. Results and Interpretation Changes

Results are reported against the four audit questions of Sec. 6.8. Unless stated otherwise, headline comparisons refer to the native-capability setting; matched stress-test results are reported separately and never redefine a method’s native behavior. Metrics computed under METRIC_SPEC v1.0, 1.1 with the RAFT flow backbone and ORB features.

7.1. Native-Capability Audit

Table 2 summarizes SNF-Bench measurements under released inference configurations at the two principal horizons, with all four reported in the supplementary material; Table 3 records each system’s resolution, frame rate, and maximum supported horizon. Unsupported horizons appear explicitly rather than being populated with modified settings.

Static fidelity. fBD captures accumulated geometric displacement; NBF captures residual instantaneous motion inside Ω_{static} . The two are not redundant: a system can hold its structure in place while its static region shimmers, and the reverse. Across the 7 audited systems at 60s the two agree on the extremes and diverge in the middle of the field, so both are reported

Method	Resolution	FPS	Max horizon
CausVid	832×480	16	240s
Self-Forcing	832×480	16	240s
Infinite-Forcing	832×480	16	240s
Rolling-Forcing	832×480	16	240s
Reward-Forcing	832×480	16	240s
LongLive	832×480	16	240s
Causal-Forcing	832×480	16	240s

Table 3. Native deployment configurations. Every system is evaluated at its released resolution and frame rate; no silent modification is made to populate an unsupported horizon.

rather than one standing for the other.

Motion persistence. Dynamic-region behavior is evaluated at the joint operating point (MCFF($\mathcal{L}$), FP), never FP alone: sustained flow requires non-trivial late motion and high retention. Reporting MCFF-E beside MCFF-L makes the degenerate case visible—a system with negligible motion throughout cannot reach a high FP without also displaying a small MCFF-E.

Static-dynamic operating regimes. Figure 3 places each system in the static-drift-surviving-flow plane, exposing qualitatively different regimes—low-drift/low-motion, high-motion/high-drift, and low-drift/non-trivial-motion—without collapsing the axes into a scalar score. The two extremes are informative in opposite directions: Infinite-Forcing attains the lowest static drift of any audited system while placing last on surviving late motion, and Causal-Forcing sits alone at the opposite corner.

7.2. Drift Leakage Changes the Interpretation of Apparent Motion

We ask whether large apparent motion reflects intended dynamic-region activity or coherent scene displacement, considering whole-frame motion scores jointly with NBF, DLR and DAR. A generic motion score answers whether substantial image motion exists; SNF-Bench asks where in the frame it originates and how much survives compensation for background displacement. A method may therefore exhibit both a high whole-frame motion score and a high DLR—not a contradiction, but a different interpretation of the same motion.

The effect is systematic rather than anecdotal. Over the audited systems at 60s, VBench Dynamic Degree correlates with SNF-Bench’s static-region flow at Spearman $\rho = +0.93$ (Table 4): the systems a generic metric reads as most dynamic are, to a close approximation, the systems whose backgrounds move most. Figure 4 shows the consequence for ordering at every

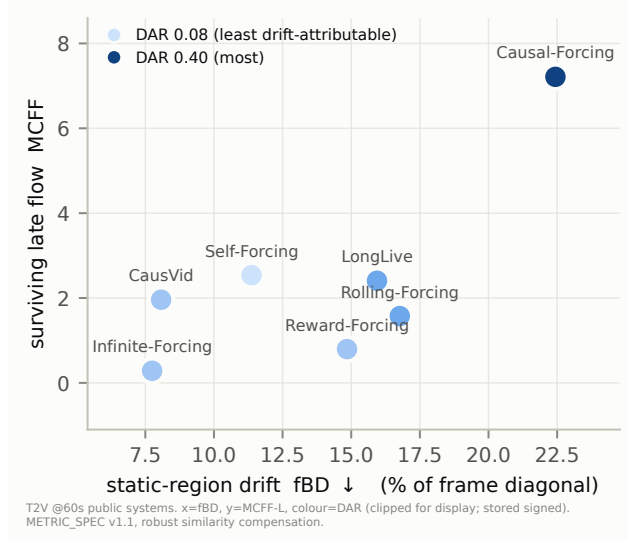


Figure 3. Static-fidelity-flow-persistence operating regimes under native deployment. Colour encodes DAR, the share of measured dynamic-region flow explained by a global similarity model. The view exposes trade-offs; it does not define a leaderboard.

generic metric	fBD	NBF	MCFF	DLR
VB-DD	+0.75	+0.93	+0.89	+0.36
VB-bg	-0.64	-0.71	-0.75	-0.50
VB-smooth	-0.96	-0.82	-0.61	-0.64
VB-flick	-0.96	-0.82	-0.61	-0.64

Table 4. Rank disagreement at 60s. Spearman correlation between generic whole-frame metrics and SNF-Bench factors over method-level means. A positive correlation between apparent motion and background flow is the effect the benchmark isolates.

horizon. Causal-Forcing ranks 1 of 7 on Dynamic Degree and 7 of 7 on NBF; Infinite-Forcing occupies the mirror position, ranking 1 on static fidelity and 7 on apparent motion. Read jointly with its magnitudes, Causal-Forcing’s DLR of 0.94 indicates static-region motion energy almost equal to that of the region intended to move, while its DAR of 0.40 indicates that a rigid global model accounts for only part of it—the remainder is non-rigid instability, which a translation-only account would have misdescribed.

Table 5 records every interpretation change meeting three pre-specified conditions: (1) generic and SNF measurements are computed on the same outputs; (2) the relevant SNF metric passed the validation of Sec. 5; and (3) the rank difference exceeds a fixed threshold rather than reflecting a small ordinal shift. The rule is applied to all public methods, so no case satisfying it

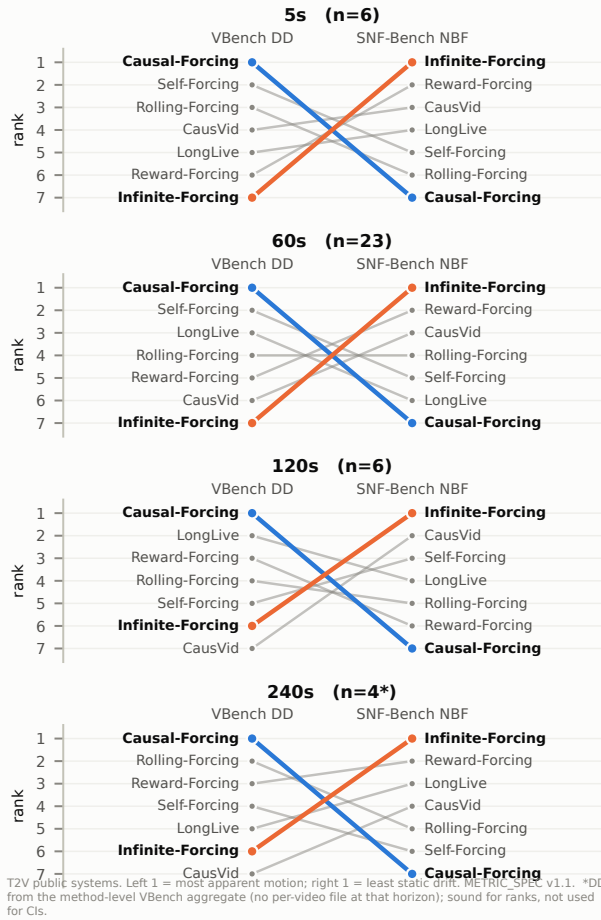


Figure 4. Rank disagreement between whole-frame motion and static-region drift, at every audited horizon. Both quantities are independent of the global-motion estimator, so the disagreement is a property of the two metrics rather than of our implementation. Short horizons carry few prompts; the claim concerns the ordering two metrics induce over the same clips, which a larger prompt set would refine but not create.

can be selectively omitted. The analysis is complementary: whole-frame metrics measure what they were designed to measure, and SNF-Bench exposes where that quantity spatially originates.

7.3. Category and Horizon Dependence

Aggregates can conceal scene dependence, so all headline numbers are macro-averaged over the 6 flow-medium categories of Sec. 3 rather than averaged over prompts directly. This matters: the 60s prompt set is 61% channel water, so a prompt-level mean is substantially a measurement of one medium. Complete per-category values appear in the supplementary material, together with the category balance of every track and

horizon.

Horizon dependence is read from the per-horizon panels of Table 2. A useful long-horizon generator should not merely score well at one duration but remain comparatively stable as the horizon grows. The 5s tier serves as an initialization sanity check; the 240s tier is exploratory and determines no principal conclusion, but it is reported because the ordering it induces is itself evidence about metric behavior under extended rollout.

7.4. Deployment Sensitivity

Native capability is finally compared against the matched stress setting via the per-checkpoint change $\Delta m = m_{\text{matched}} - m_{\text{native}}$ for the checkpoints evaluated under both regimes; absolute matched scores are not interpreted as the published method’s performance. Table 6 reports representative factors. Large changes indicate sensitivity to temporal block size, cache policy, or denoising schedule; small changes indicate that static-dynamic behavior transfers across deployment regimes. The audit establishes that behavior changes under the matched regime; attributing a change to a specific architectural mechanism requires method-specific intervention beyond SNF-Bench’s scope.

7.5. Summary of Audit Findings

Answering the four pre-specified questions of Sec. 6.8:

1. Static fidelity. Audited systems separate cleanly on fBD and NBF, and the separation widens with horizon. No system holds its static support fixed while also sustaining dynamic-region motion at the level it produces early in the rollout.
 2. Motion persistence. Read jointly, MCFF-E, MCFF-L and FP distinguish sustained flow from two different degenerate cases—motion that decays, and motion that was never present—which FP alone conflates.
 3. Metric interpretation. Whole-frame motion and static-region drift induce near-opposite orderings ($\rho = +0.93$ against NBF), and the inversion is stable across horizons. This is the benchmark’s principal finding.
 4. Deployment sensitivity. Static-dynamic behavior is configuration-dependent, so a single number attached to a method name is under-specified without its deployment regime.
- No result subsection is added or removed based on which system performs best.

8. Limitations and Conclusion

Scope. SNF-Bench is intentionally narrow: long-horizon fixed-camera nature-flow generation, where

Method	Generic evidence	SNF-Bench evidence	Interpretation
CausVid	VB-DD rank 6/7	NBF rank 3/7, MCFF-L rank 4/7	stable sequence with decaying intended motion
Causal-Forcing	VB-DD rank 1/7	NBF rank 7/7, MCFF-L rank 1/7	substantial apparent motion, strongly contaminated by background displacement
Infinite-Forcing	VB-DD rank 7/7	NBF rank 1/7, MCFF-L rank 7/7	stable sequence with decaying intended motion
LongLive	VB-DD rank 3/7	NBF rank 6/7, MCFF-L rank 3/7	substantial apparent motion, strongly contaminated by background displacement
Reward-Forcing	VB-DD rank 5/7	NBF rank 2/7, MCFF-L rank 6/7	stable sequence with decaying intended motion
Self-Forcing	VB-DD rank 2/7	NBF rank 5/7, MCFF-L rank 2/7	substantial apparent motion, strongly contaminated by background displacement

Table 5. Interpretation changes between generic whole-frame metrics and SNF-Bench at 60s. Neither evaluation is labelled correct; the table identifies information hidden by whole-frame aggregation. Cases are selected by a fixed rank-gap rule applied to all public methods, so none can be selectively omitted.

Checkpoint	horizon	ΔfBD	ΔNBF	$\Delta MCFF-L$	ΔD_{opt}	Optical-flow and feature estimators under severe degradation; controlled validation establishes their behavior over the benchmark’s operating range, and they should be treated as calibrated diagnostic proxies, not exact physical decompositions.
Causal-Forcing++ (2-step)	5s	+2.47	-0.61	-0.18	-0.04	972
Causal-Forcing++ (2-step)	60s	-0.76	-0.95	-0.15	-0.08	973
Causal-Forcing++ (2-step)	120s	-1.71	-8.46	-0.61	-0.17	974
Causal-Forcing++ (2-step)	240s	+4.06	+0.88	-0.06	-0.23	975
Causal-Forcing++ (1-step)	60s	-5.05	+1.80	+0.13	-0.20	976
Causal-Forcing++ (1-step)	120s	-6.34	+2.00	+0.41	-0.14	

Table 6. Deployment sensitivity. Change from each checkpoint’s native configuration to the common long-horizon wrapper, for the two checkpoints evaluated both ways. Absolute matched scores are never read as the published method’s performance.

static support and dynamic flow can be meaningfully separated. It does not rank general-purpose generation across moving-camera cinematography, articulated action, narrative scene change, or interactive simulation, and results should be read as evidence about this controlled regime.

Persistence is not realism. The benchmark measures whether motion occurs in expected regions, remains non-trivial over time, and is uncontaminated by background displacement. It does not determine whether water, smoke, or fire obey physically correct dynamics: a sequence can be drift-free and persistent while remaining implausible or repetitive. Human perceptual evaluation and physics-aware diagnostics are complementary, not superseded.

Masks and estimators are approximate. The static-flow partition is imperfect near reflections, spray, foliage, and smoke boundaries; method-independent construction, agreement measurement, transition bands, and erosion-dilation analysis bound this uncertainty but do not remove it, and conclusions unstable under mask perturbation are not reported as headline claims. Likewise, all measurements inherit the failure modes of

Two settings, two questions. The native audit preserves each released system’s deployment regime because that regime is part of the system; the matched stress test deliberately perturbs it. The settings must not be merged into one leaderboard, and matched-setting degradation must not be attributed to a specific architectural cause without intervention beyond this benchmark’s scope.

Conclusion. We introduced SNF-Bench (Static-Natural Flow), an evaluation framework for long-horizon fixed-camera video generation built on one distinction: some regions of a scene must hold still, and others must not stop moving. The framework measures Static Fidelity, Flow Persistence with absolute magnitude, and Drift Leakage; validates every measurement against known corruptions before any model comparison; and audits public systems under both their native configurations and a matched stress test, separating released-system capability from deployment sensitivity. Whole-frame metrics answer whether a video moves and whether it is consistent. SNF-Bench answers the question those aggregates cannot: where the motion comes from, and whether it lasts. As generation horizons extend from seconds to minutes, that spatial accounting becomes the difference between a scene that flows and a scene that merely drifts—and it is precisely what long-horizon evaluation now needs to measure.

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